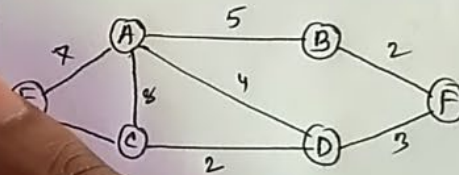
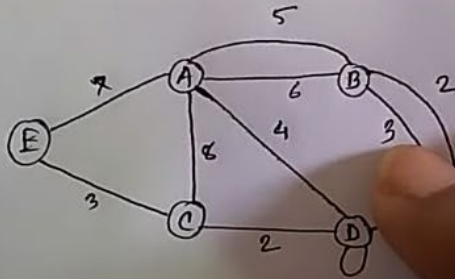


Kruskal's Algorithm for MST

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⇒

- (i) Graph হতে Loop delete করা হবে।
- (ii) Graph হতে parallel edge delete করা হবে।
- (iii) Cycle নেই কিনা পরীক্ষা করা হবে, নইলে MST গঠন করা হবে।



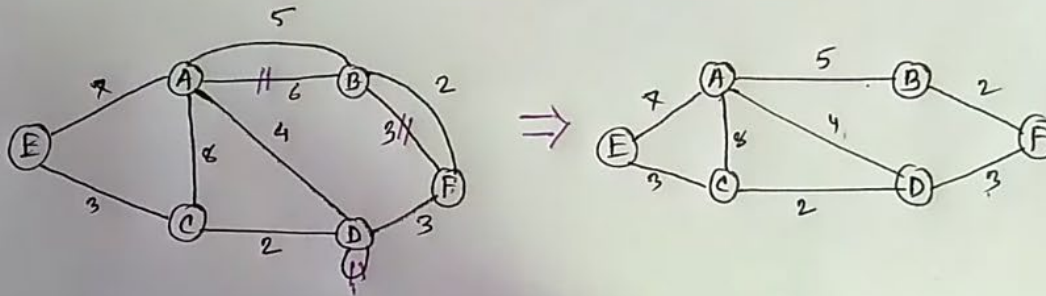
for MST

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⇒

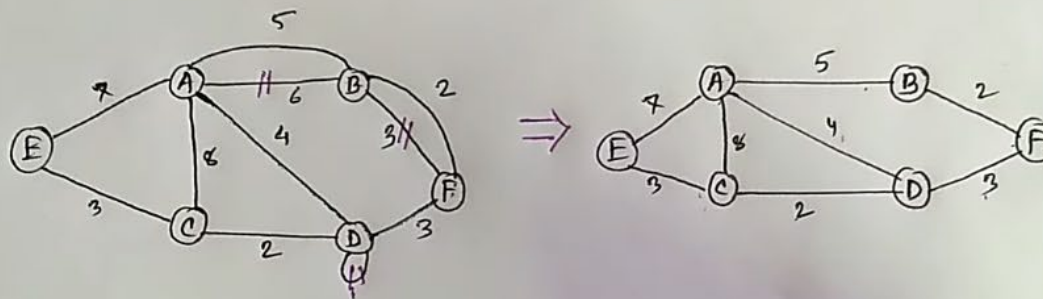
- (i) Graph ରୁ Loop delete କରାଯାଏ।
- (ii) Graph ରୁ parallel edge delete କରାଯାଏ।
- (iii) Cycle ସୃଷ୍ଟି ହେଉ ନା, ନହେଉ MST ଏବଂ ଅଟେ।



When deleting parallel edges,
we keep the lower weighted
edge

⇒

- ① Graph হতে loop delete করা হবে।
- ② Graph হতে parallel edge delete করা হবে।
- ③ Cycle তৈরি হলে না, তবে MST গঠন করা হবে।

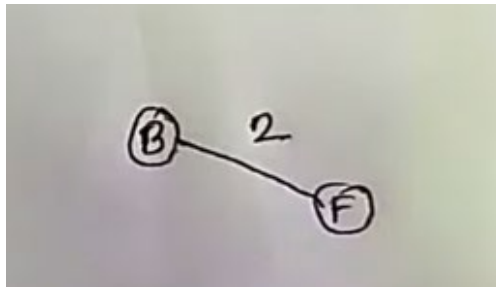


BF	→ 2
CD	→ 2
DF	→ 3
CE	→ 3
AD	→ 4
AB	→ 5
AE	→ 7
AC	→ 8

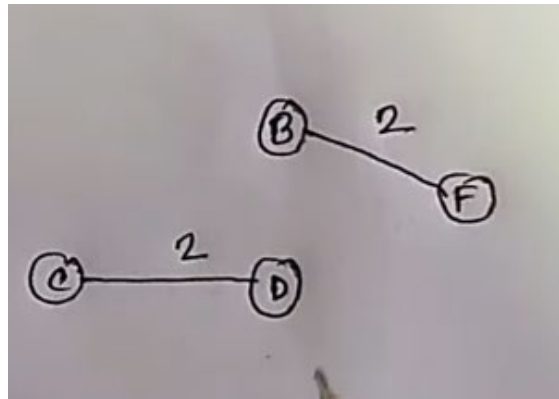
We will sort the edges with the increasing order of their weight

We can create this table for convenience

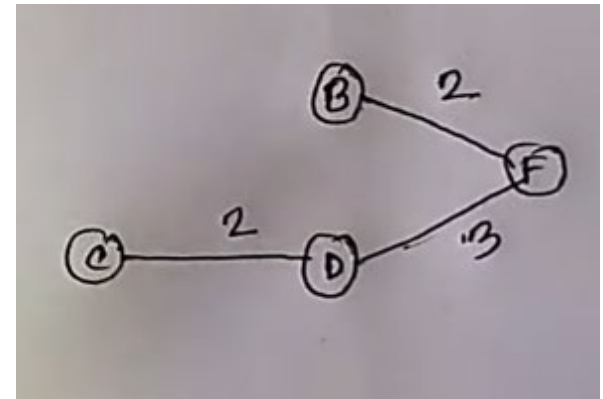
Step 1



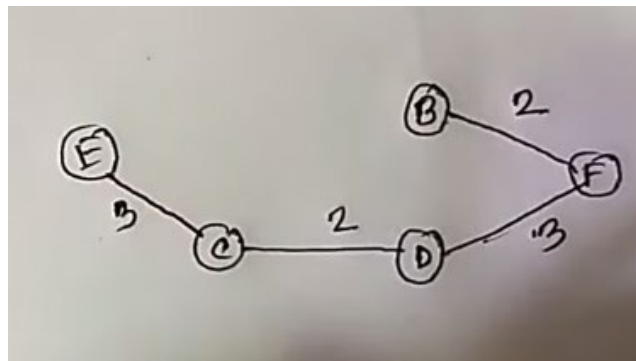
Step 2



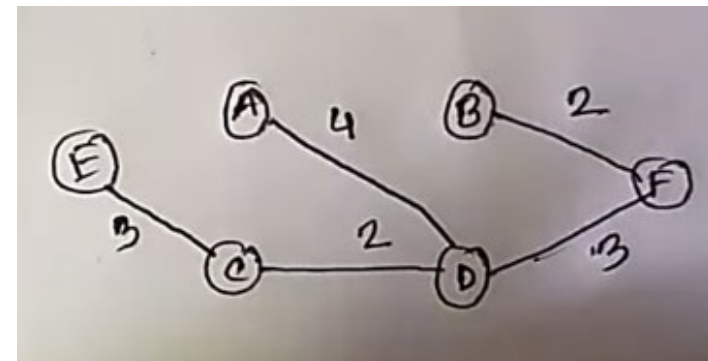
Step 3



Step 4

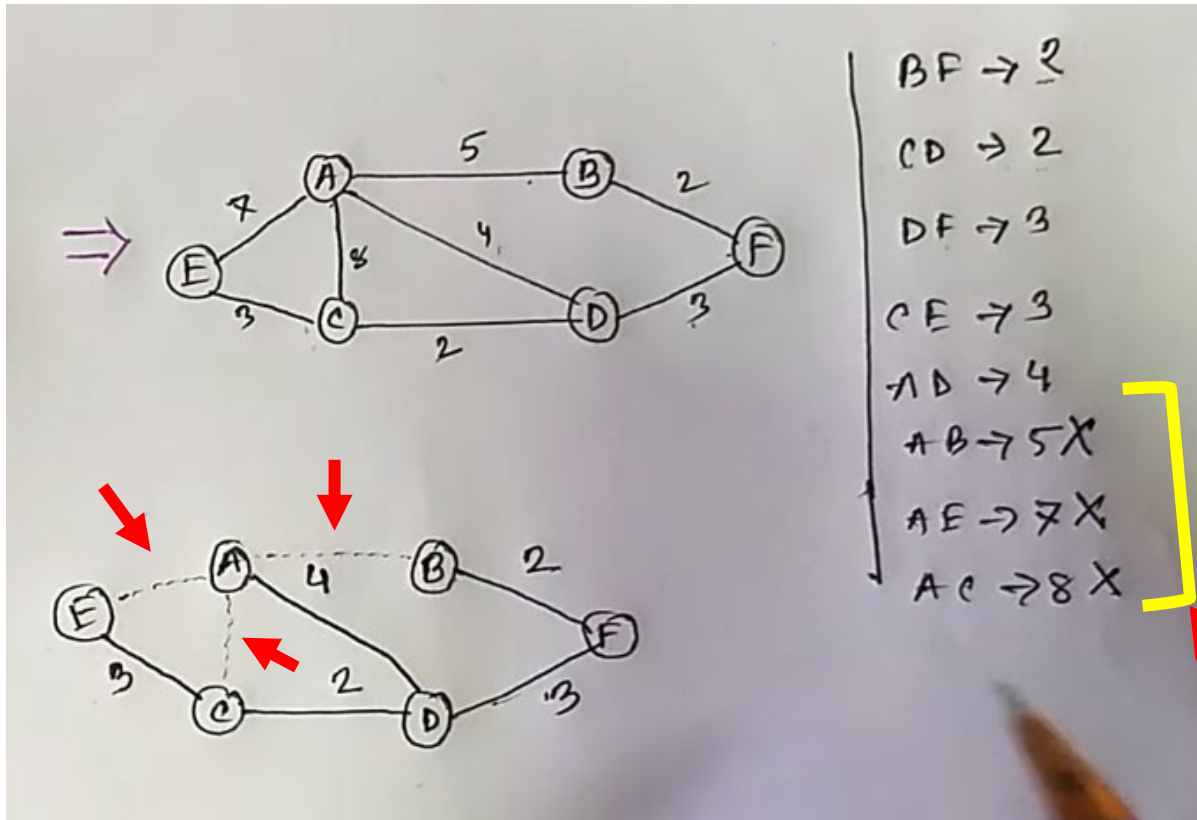


Step 5



BF $\rightarrow 2$
 CD $\rightarrow 2$
 DF $\rightarrow 3$
 CE $\rightarrow 3$
 AD $\rightarrow 4$
 AB $\rightarrow 5$
 AE $\rightarrow 7$
 AC $\rightarrow 8$

Step 6

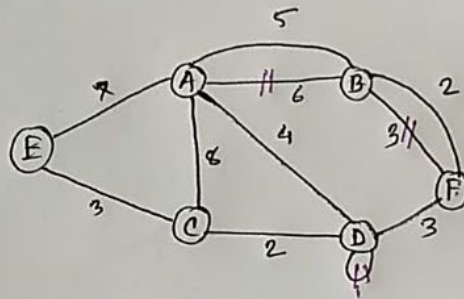


"Greedy Method"

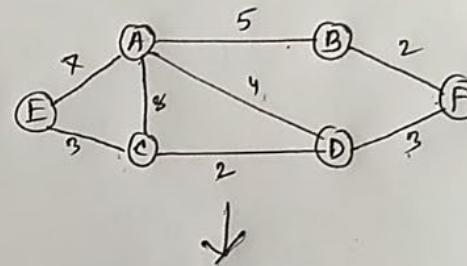
We can get MST without finding the number of Spanning Trees

⇒

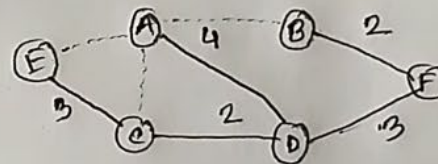
- (i) Graph ਤੋਂ Loop delete ਕਰੋ।
- (ii) Graph ਤੋਂ parallel edge delete ਕਰੋ।
- (iii) Cycle ਰੱਖਣ ਵਾਲੇ ਜਾਂ, ਜੇਕਰ MST ਦੀ ਤਰ੍ਹਾਂ



⇒



↓



$BF \rightarrow 2$
 $CD \rightarrow 2$
 $DE \rightarrow 3$
 $CE \rightarrow 3$
 $AD \rightarrow 4$
 $AB \rightarrow 5X$
 $AE \rightarrow 7X$
 $AC \rightarrow 8X$

Since it follows the properties of MST,

- Connected
- $E' = |V| - 1$
- $V = V'$

This is a valid MST

$G(V, E)$
 $G'(V', E')$
 $V' = V$
 $E' = |V| - 1$

